Convex Roof C_GME — Hubbard L=4 Ne=4 t=1.0 U=[[0.70089271 -0.64932977 -0.27994796 -0.09353822]
[0.09353823 0.27994797 -0.64932978 0.7008927]
[0.70089272 0.64932976 0.27994795 -0.09353824]
[0.09353823 -0.27994794 0.64932975 0.70089274]] (3-body mixed RDM)

```
System:  L=4  Ne=4  t=1.0  U=[[ 0.70089271 -0.64932977 -0.27994796 -0.09353822]
 [ 0.09353823  0.27994797 -0.64932978  0.7008927 ]
 [ 0.70089272  0.64932976  0.27994795 -0.09353824]
 [ 0.09353823 -0.27994794  0.64932975  0.70089274]]  mode=uniform
RDM:  3-body  sites=(0, 1, 2)  dim=8x8

============================================================

  Tr(rho)              = 1.00000000
  Tr(rho^2)            = 0.46262004  (mixed state)
  Rank(rho)            = 4

============================================================

  Method: minimize  sum_i p_i C_GME(|psi_i>)
  over all decompositions  rho = sum_i p_i |psi_i><psi_i|
  via unitary mixing:  W = V sqrt(D) U†
  Optimizer: L-BFGS-B  with 200 random restarts

============================================================

  C_UB  (eigendecomp upper bound) = 0.71636717
  C_num (existing numerical)      = 0.50351431
  C_GME (unitary optimization)    = 0.50351431
  Improvement over C_num          = 1.26e-13

============================================================

  Best decomposition  rho = sum_i p_i |psi_i><psi_i|:

    i=0:  p_i=0.250000   C_GME(psi_i)=0.503514   p_i * C_GME = 0.125879
    i=1:  p_i=0.250000   C_GME(psi_i)=0.503514   p_i * C_GME = 0.125879
    i=2:  p_i=0.250000   C_GME(psi_i)=0.503514   p_i * C_GME = 0.125879
    i=3:  p_i=0.250000   C_GME(psi_i)=0.503514   p_i * C_GME = 0.125879

  Sum  p_i * C_GME(psi_i) = 0.50351431
```